

DFRL (Layer 2): Dynamic Failure Risk Learning

OFRS: Online Failure Risk Scorer

Reorders queue by smoothed per-dimension risk

update statistics

sort by $r(c)$

Queue Q_t

pop lowest-risk

HLS Synthesis
(Bambu / Dynamatic)

Outcome
(success / fail-type)

extract signatures

RPE: Recurrent Pattern Extractor

Extracts failure signatures

skip matched configs

initialize Q

SCF (Layer 1): Static Constraint Filter

Rule-based, zero false-prune

Partition of C

$C_{blocked}$
(removed)

$C_{suppressed}$
(deferred)

C_{active}
(kept)

filter

Config space C
(N=420 for Bambu,
192 for Dynamatic)

Evidence hierarchy: SCF (proven) $\rightarrow$ permanent block | RPE (typed empirical) $\rightarrow$ hard skip |
OFRS (marginal empirical) $\rightarrow$ reorder only

